BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Validation and Authentication

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. After correcting an assessment answer, why attempt a new question?

2. After one ascending bubble-sort pass on [7, 3, 6, 2], what is the list?
   A) [2, 3, 6, 7]
   B) [3, 6, 2, 7]
   C) [7, 3, 6, 2]
   D) [2, 6, 3, 7]

3. "An IF statement must execute every possible branch regardless of its condition." Correct this. Explain why.

4. A circuit has X = A AND B, followed by Q = X OR C. For input rows 000, 011, 110, 111, give the pairs (X,Q). Why is a column for X useful?

Today's objectives

Explain the core ideas of validation and authentication using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Validation
Checks whether input meets a defined rule, not whether it is factually true.

Range check
Rejects a number outside stated minimum and maximum boundaries.

Presence and type
Detects missing required input or wrong type/format.

Authentication
Checks a user's claimed identity, which differs from validating an entered value.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Validation
Checks whether input meets a defined rule, not whether it is factually true.

Range check
Rejects a number outside stated minimum and maximum boundaries.

Presence and type
Detects missing required input or wrong type/format.

Authentication
Checks a user's claimed identity, which differs from validating an entered value.

I Do - Worked model

Design a range check for an integer mark from 0 to 100 inclusive.

- Input the candidate mark.
- Test mark >= 0 AND mark <= 100.
- Accept only when both conditions are true.
- Otherwise display an error and ask for a new input.

We Do - Guided practice

1. A form accepts ages from 13 to 18. Write an acceptance condition.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Finish the inclusive range check for one IPv4 octet.

```
octet = 255
if ... then
print("Valid range")
else
print("Invalid")
endif
```

You Do - Independent exam questions

Q2. Explain validation versus verification. [2 marks]

Q3. Give a suitable boundary value when valid ages are 13-18. [2 marks]

Q4. State one difference between authentication and validation. [2 marks]

Exit Ticket - True or False?

Validation proves a pupil's entered date is factually correct.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. To check that the method can be applied independently rather than copied.

2. B - [3, 6, 2, 7]

3. It chooses a branch using a condition. Selection can be nested.

4. (a) (0,0), (0,1), (1,1), (1,1). X shows the first gate's output. This helps work out Q and find errors.

I Do - Explain the model

Input the candidate mark.

Test mark >= 0 AND mark <= 100.

Accept only when both conditions are true.

Otherwise display an error and ask for a new input.

We Do - Model answers

1. age >= 13 AND age <= 18.

Code First - Answer

```
octet = 255
if octet >= 0 AND octet <= 255 then
print("Valid range")
else
print("Invalid")
endif
```

Expected output: Valid range

You Do - Self-mark

Q2: Validation checks a rule or format; verification checks data matches the original/source input.

Q3: 13 or 18, plus an adjacent invalid such as 12 or 19 for a test set.

Q4: Authentication confirms an asserted identity; validation checks input against constraints.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.